

Group-wise PCA of SH Coefficients for 3DGS Compression

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Abstract

MPEG is currently standardizing Gaussian Splat Coding (GSC) as a compression standard for 3D Gaussian Splatting (3DGS) models. In this process, a video-based compression framework, referred to as Test Model Candidate Video (TMCV), is used to efficiently compress Gaussian attributes, including spherical harmonics (SH) coefficients. This paper proposes a group-wise Principal Component Analysis (PCA) approach to improve the compression efficiency of SH coefficients within the TMCV framework. The proposed method groups SH coefficients based on their correlation characteristics and applies PCA individually to each group, thereby enabling more effective exploitation of the structural correlation properties inherent in SH coefficients. Experimental results demonstrate that the proposed method outperforms the TMCV anchor in terms of compression efficiency, achieving BD-rate reductions of 19.0%, 19.5%, and 17.4% for RGB-PSNR, YUV-PSNR, and YUV-SSIM, respectively.

Index Terms: 3D Gaussian Splatting (3DGS), Gaussian Splat Coding (GSC), Spherical Harmonics, Principal Component Analysis (PCA), TMCV.

1 INTRODUCTION

3D Gaussian Splatting (3DGS) [1] has recently emerged as an efficient representation for real-time radiance field rendering. However, the large number of Gaussian attributes poses significant challenges in terms of storage and transmission, making compression essential for practical deployment.

MPEG is currently standardizing Gaussian Splat Coding (GSC) [2], which adopts a video-based compression framework known as the Test Model

Candidate Video (TMCV) [3, 4] to efficiently encode Gaussian attributes, including spherical harmonics (SH) coefficients. Due to their high dimensionality and strong inter-coefficient correlations, SH coefficients account for a substantial portion of the overall bitrate.

This paper proposes a grouped principal component analysis (PCA) based preprocessing method for SH coefficients within the TMCV framework. The proposed method groups SH coefficients according to their correlation characteristics and applies PCA independently to each group, enabling more effective redundancy reduction while preserving the structural properties of SH coefficients.

Experimental results demonstrate that the proposed group-wise PCA-based preprocessing method achieves improved compression efficiency compared to the TMCV anchor, yielding consistent BD-rate reductions in terms of RGB-PSNR, YUV-PSNR, and YUV-SSIM. These results confirm the effectiveness of the proposed method in enhancing the compression performance of SH coefficients in 3DGS.

2 BACKGROUND

2.1 TMCV COMPRESSION FRAMEWORK

The TMCV framework adopts a video-based compression approach to encode Gaussian attributes in GSC. In TMCV, Gaussian parameters are first sorted and packed into a pseudo-video sequence through PLAS [5] sorting and attribute packing. The packed attributes are then quantized and encoded using standard video codecs, as illustrated in Fig 1.

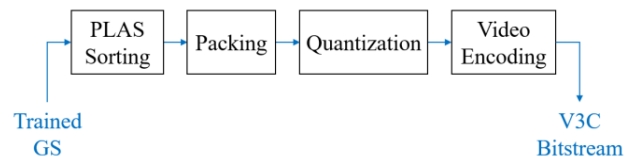


Figure 1: TMCV encoding pipeline.

Within this framework, SH coefficients are treated as high-dimensional attributes and constitute a significant portion of the overall bitrate. Due to their dimensionality and structured correlation characteristics, efficient representation of SH coefficients is critical for improving compression performance. However, these characteristics are not explicitly considered in the baseline TMCV framework,

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which may limit compression efficiency.

2.2 SH REPRESENTATION IN 3DGS

3D Gaussian Splatting (3DGS) represents scene appearance using a collection of 3D Gaussian primitives, where each primitive encodes color and view-dependent radiance through spherical harmonics (SH). SH coefficients are defined according to the degree and order (l, m) of the basis functions and are commonly used to approximate angular radiance.

The view-dependent color $C(v)$ for a given view direction v can be expressed as

$$C(v) = \sum_{l=0}^L \sum_{m=-l}^l c_{l,m} Y_l^m(v) \quad (1)$$

Where $Y_l^m(\cdot)$ denotes the SH basis functions and $c_{l,m}$ represents the corresponding SH coefficients to be encoded.

In typical 3DGS configurations, SH coefficients up to degree $l = 3$ are employed, resulting in a total of 45 coefficients per Gaussian primitive across RGB color channels.

However, the relatively high dimensionality of SH coefficients makes them a major contributor to the overall bitrate in 3DGS compression. Moreover, due to the mathematical structure of spherical harmonics, SH coefficients exhibit non-uniform correlation characteristics, which motivates the need for more effective preprocessing and redundancy reduction techniques.

2.3 CORRELATION OF SH COEFFICIENTS

Due to the mathematical structure of spherical harmonics, SH coefficients exhibit non-uniform correlation characteristics.

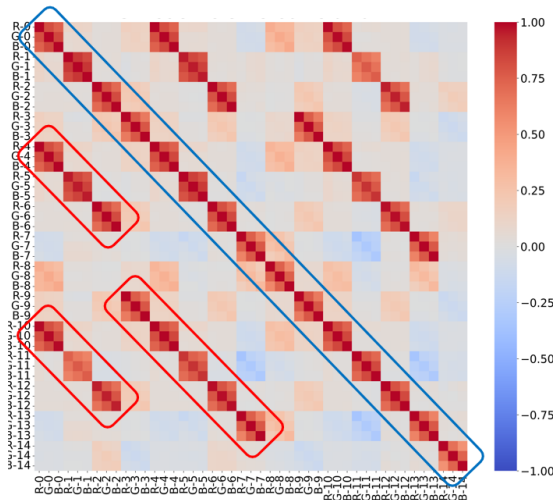


Figure 2: Correlation heatmap of SH coefficients for the Bartender sequence.

In particular, coefficients sharing the same order m tend to show strong correlations across color channels (R, G, and B), while correlations across different orders or degrees are relatively weaker.

Fig. 2 shows a correlation heatmap of SH coefficients extracted from the Bartender sequence. As observed in the figure, clusters of highly correlated coefficients are clearly formed, especially among coefficients with the same order m , indicating that SH coefficients possess localized correlation patterns rather than uniform global correlations. These observations suggest that treating all SH coefficients as a single high-dimensional vector may not be optimal for compression, as localized correlation structures can be obscured when global statistical models are applied.

3 PROPOSED METHOD

Motivated by the localized correlation characteristics of SH coefficients discussed in Section 2.2, this paper proposes a grouped PCA-based preprocessing method in which PCA is applied independently to correlation-aware SH coefficient groups.

The proposed method is integrated into the TMCV compression framework as a preprocessing step prior to quantization and video encoding. Importantly, it does not require any modification to the core TMCV coding tools, ensuring full compatibility with existing implementations.

3.1 PRINCIPAL COMPONENT ANALYSIS

Principal component analysis (PCA) is a statistical technique that projects high-dimensional data onto a lower-dimensional subspace through an orthogonal linear transformation while preserving the maximum possible variance of the original data. By retaining the most significant principal components, PCA enables effective dimensionality reduction with minimal information loss.

In addition to dimensionality reduction, PCA reduces redundancy by decorrelating variables that exhibit strong statistical correlation. This property is particularly beneficial for compression, as it concentrates signal energy into a smaller number of components, facilitating more efficient quantization and entropy coding.

In the proposed method, PCA is applied independently to each SH coefficient group defined in Section 3.2. . For each group, the coefficients are normalized by removing the mean and scaling by the standard deviation, and then projected onto a reduced PCA subspace that preserves a predefined explained variance ratio. This group-wise PCA strategy captures localized correlation structures within each group while avoiding unnecessary mixing of weakly correlated coefficients, forming an effective preprocessing step for SH coefficient compression.

3.2 SH COEFFICIENT GROUPING STRATEGY

In the proposed method, SH coefficients are first partitioned into multiple groups according to their degree and order (l, m). Coefficients sharing the same order m are grouped together, as they exhibit strong inter-channel correlations across the R, G, and B color components.

For SH coefficients up to degree $l = 3$, a total of 45 coefficients are divided into seven groups, each corresponding to a specific order m . This structured grouping enables more effective exploitation of correlation characteristics in subsequent processing stages, forming the basis for correlation-aware reduction.

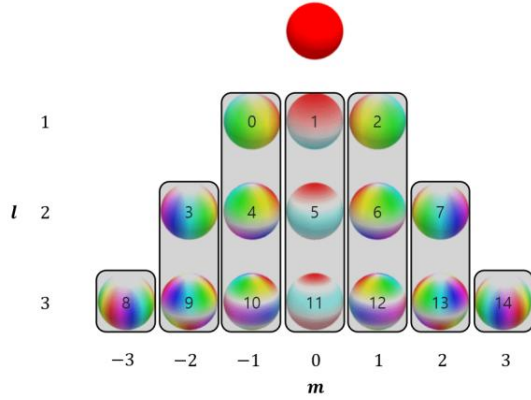


Figure 3: Grouping strategy of SH coefficients based on degree l and order m .

3.3 INTEGRATION WITH THE TMCV PIPELINE

The proposed grouped PCA-based preprocessing method is seamlessly integrated into the TMCV compression pipeline. As illustrated in Fig. 3, the SH coefficient grouping and group-wise PCA operations are inserted as a preprocessing stage prior to quantization and video encoding.

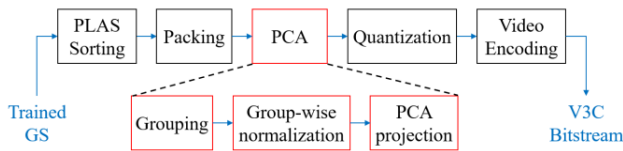


Figure 4: TMCV encoding pipeline with the proposed grouped PCA-based preprocessing method.

After group-wise PCA transformation, the coefficients are quantized and encoded using the standard TMCV framework without any modification to the downstream coding modules. This design ensures full compatibility with existing TMCV implementations while enabling improved compression efficiency through correlation-aware preprocessing.

4 EXPERIMENT RESULTS

The proposed grouped PCA-based preprocessing method was implemented within the TMCV framework and evaluated under the common test conditions (CTC) [6] defined for GSC.

All experiments were conducted using identical encoding configurations to ensure fair comparison. Quantization parameters for Gaussian attributes, including position, opacity, scale, rotation, and SH coefficients, followed the CTC settings.

For the proposed method, the explained variance ratio of PCA was fixed to 99%. Rate-distortion performance was evaluated using BD-rate metrics in terms of RGB-PSNR, YUV-PSNR, and YUV-SSIM, with the TMCV anchor configuration serving as the reference.

Table 2 and Table 3 summarize the rate-distortion performance of the proposed group-wise PCA-based preprocessing method compared to the TMCV anchor configuration.

Table 1: Quantization parameter (QP) settings for Gaussian attributes under the GSC CTC

RP	Position	Opacity	Scale	Rotation	SH DC	SH AC
1	0	7	7	2	7	7
2	0	17	7	2	7	17
3	0	17	12	7	7	22
4	0	22	12	17	7	32

Table 2: 1F Forward facing sequences

Sequence	BD-rate RGB-PSNR	BD-rate YUV-PSNR	BD-rate YUV-SSIM
bartender_semitracked	-20.8%	-21.0%	-18.5%
cinema_semitracked	-15.9%	-15.9%	-13.6%
breakfast_semitracked	-25.3%	-26.3%	-23.8%
Breakfast_untracked	-9.1%	-9.5%	-8.5%
bartender_tracked	-20.7%	-21.3%	-19.1%
cinema_tracked	-16.6%	-16.8%	-14.8%
breakfast_tracked	-24.8%	-25.8%	-23.7%
Average	-19.0%	-19.5%	-17.4%

Table 3: 1F Object-centric sequences

Sequence	BD-rate RGB-PSNR	BD-rate YUV-PSNR	BD-rate YUV-SSIM
manwithfruit_tracked	-10.7%	-11.4%	-14.9%
lego_ferrari	-9.5%	-11.2%	-14.9%
lego_bugatti	-19.5%	-20.5%	-26.8%
cricket_player	-10.5%	-12.4%	-15.6%
plant	-18.1%	-19.7%	-28.8%
solo_tango_female	-4.7%	-8.3%	-10.9%
solo_tango_male	-9.2%	-12.6%	-11.6%
tango_duo	-13.9%	-16.6%	-20.6%
tennis_player	-6.9%	-9.8%	-12.2%
flowerdance	-6.5%	-7.3%	-9.1%
gymnast	-10.2%	-1.9%	2.0%
Average	-10.0%	-12.0%	-14.9%

For forward-facing sequences, the proposed method achieves an average BD-rate reduction of 19.0% in RGB-PSNR, 19.5% in YUV-PSNR, and 17.4% in YUV-SSIM relative to the anchor. Consistent bitrate savings are observed across both tracked and semitracked configurations, indicating that the proposed preprocessing effectively reduces redundancy in SH coefficients.

For object-centric sequences, the proposed method also demonstrates improved compression efficiency, achieving average BD-rate reductions of 10.0% in RGB-PSNR, 12.0% in YUV-PSNR, and 14.9% in YUV-SSIM compared to the anchor. Although the performance gain varies depending on sequence characteristics, the overall results confirm that the proposed method consistently improves rate-distortion performance.

These results indicate that correlation-aware preprocessing of SH coefficients can significantly enhance compression efficiency in the TMCV framework without modifying the core coding tools.

5 CONCLUSION

This paper proposes a group-wise PCA-based method for SH coefficients to enable efficient 3DGS compression. The proposed method is implemented as a preprocessing technique within the TMCV framework of GSC. Experimental results confirm that the proposed method achieves significant BD-rate reductions compared to the anchor across both forward-facing and object-centric sequences. For some object-centric sequences, relatively smaller performance gains are observed, which can be attributed to the dispersion of SH coefficient correlations and diverse viewing conditions.

As future work, it is necessary to analyze the trade-offs between global PCA and group-wise PCA approaches in terms of BD-rate performance and computational complexity. Based on this analysis, it is required to select more effective PCA strategies according to content characteristics.

Acknowledgments

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References

[1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D Gaussian splatting for real-time radiance field rendering,” *ACM Trans. Graphics*, vol. 42, no. 4, pp. 1–14, 2023.

[2] ISO/IEC JTC 1/SC 29/WG 04, “AHGs established at the 17th meeting,” Doc. N0568, Kemer, Türkiye, Nov. 2024.

[3] ISO/IEC JTC 1/SC 29/WG 04, “Description on joint exploration experiments 6: 3D Gaussian representation and compression,” Doc. N0560, Sapporo, Japan, Jul. 2024.

[4] GSC TMCV software, Git repository, <https://git.mpeg.expert/MPEG/Explorations/GSC/gsc-software/mpeg-gsc-tmcv>, accessed Jan. 24, 2026.

[5] ISO/IEC JTC 1/SC 29/WG 04, “Draft common test conditions for Gaussian splat coding,” Doc. N1292, Daejeon, Korea, Jul. 2025.

[6] W. Morgenstern, F. Barthel, A. Hilsmann, and P. Eisert, “Compact 3D Scene Representation via Self-Organizing Gaussian Grids,” in *Proc. European Conf. Comput. Vision (ECCV)*, LNCS, Springer, pp. 18–34, 2024.